Web .net cie 2 task

TASK A: Hospital Appointment Management System

```
HospitalAppointment
│
├── Controllers
│   └── HomeController.cs
│
├── Models
│   ├── Doctor.cs
│   └── ErrorViewModel.cs       │
├── Extensions
│   └── DoctorExtensions.cs
│
├── Views
│   ├── Home
│   │   ├── Index.cshtml
│   │   ├── Details.cshtml
│   │   └── Posts.cshtml
│   │
│   ├── Shared
│   │   └── Error.cshtml
│   │
│   └── _ViewImports.cshtml
│
├── wwwroot
├── Program.cs
└── HospitalAppointment.csproj
```

Models/Doctor.cs

```
namespace HospitalAppointment.Models
{
    public class Doctor
    {
        public int DoctorId { get; set; }

        public string DoctorName { get; set; } = "";

        public string Specialization { get; set; } = "";

        public int Experience { get; set; }

        public decimal ConsultationFee { get; set; }
    }
}
```

Web .net cie 2 task

Models/Post.cs

```
namespace HospitalAppointment.Models
{
    public class Post
    {
        public int Id { get; set; }

        public string Title { get; set; } = "";
    }
}
```

Models/ErrorViewModel.cs

```
namespace HospitalAppointment.Models
{
    public class ErrorViewModel
    {
        public string? RequestId { get; set; }

        public bool ShowRequestId =>
            !string.IsNullOrEmpty(RequestId);
    }
}
```

Extensions/DoctorExtensions.cs

```
using HospitalAppointment.Models;

namespace HospitalAppointment.Extensions
{
    public static class DoctorExtensions
    {
        public static decimal CalculateTotalCharge(this Doctor doctor)
        {
            return doctor.ConsultationFee + 100;
        }
    }
}
```

Program.cs

```
var builder = WebApplication.CreateBuilder(args);

// MVC
builder.Services.AddControllersWithViews();

// Session
builder.Services.AddSession();
```

Web .net cie 2 task

```
// Memory Cache
builder.Services.AddMemoryCache();

// HttpClient
builder.Services.AddHttpClient();

// Response Caching
builder.Services.AddResponseCaching();

var app = builder.Build();

if (!app.Environment.IsDevelopment())
{
    app.UseExceptionHandler("/Home/Error");
}

app.UseHttpsRedirection();

app.UseStaticFiles();

// Response Cache
app.UseResponseCaching();

// Session
app.UseSession();

app.UseRouting();

app.UseAuthorization();

app.MapControllerRoute(
    name: "default",
    pattern: "{controller=Home}/{action=Index}/{id?}");

app.Run();
```

Controllers/HomeController.cs

```
using HospitalAppointment.Extensions;
using HospitalAppointment.Models;
using Microsoft.AspNetCore.Mvc;
using Microsoft.Extensions.Caching.Memory;
using System.Text.Json;

namespace HospitalAppointment.Controllers
{
    public class HomeController : Controller
    {
```

Web .net cie 2 task

```
private readonly IMemoryCache _cache;
private readonly IHttpClientFactory _httpClientFactory;

public HomeController(
    IMemoryCache cache,
    IHttpClientFactory httpClientFactory)
{
    _cache = cache;
    _httpClientFactory = httpClientFactory;
}

// Doctor List
public IActionResult Index()
{
    // Cookie - Store patient name
    Response.Cookies.Append("PatientName", "Patient");

    // Session - Store selected specialization
    HttpContext.Session.SetString(
        "Specialization",
        "Cardiology"
    );

    // IMemoryCache - 3 minutes
    var doctors = _cache.GetOrCreate("DoctorList", entry =>
    {
        entry.AbsoluteExpirationRelativeToNow =
            TimeSpan.FromMinutes(3);

        return new List<Doctor>
        {
            new Doctor
            {
                DoctorId = 1,
                DoctorName = "Dr. Amit Patel",
                Specialization = "Cardiology",
                Experience = 10,
                ConsultationFee = 500
            },

            new Doctor
            {
                DoctorId = 2,
                DoctorName = "Dr. Neha Shah",
                Specialization = "Dermatology",
                Experience = 7,
                ConsultationFee = 400
            },
```

```
            new Doctor
            {
                DoctorId = 3,
                DoctorName = "Dr. Raj Mehta",
                Specialization = "Neurology",
                Experience = 12,
                ConsultationFee = 700
            }
        };
    });

    return View(doctors);
}

// Doctor Details
[ResponseCache(Duration = 60)]
public IActionResult Details(int id)
{
    var doctors = new List<Doctor>
    {
        new Doctor
        {
            DoctorId = 1,
            DoctorName = "Dr. Amit Patel",
            Specialization = "Cardiology",
            Experience = 10,
            ConsultationFee = 500
        },

        new Doctor
        {
            DoctorId = 2,
            DoctorName = "Dr. Neha Shah",
            Specialization = "Dermatology",
            Experience = 7,
            ConsultationFee = 400
        },

        new Doctor
        {
            DoctorId = 3,
            DoctorName = "Dr. Raj Mehta",
            Specialization = "Neurology",
            Experience = 12,
            ConsultationFee = 700
        }
    };
```

```
        var doctor = doctors.FirstOrDefault(
            d => d.DoctorId == id
        );

        if (doctor == null)
        {
            return NotFound();
        }

        return View(doctor);
    }

    // Asynchronous API Request
    public async Task<IActionResult> Posts()
    {
        var client = _httpClientFactory.CreateClient();

        var response = await client.GetAsync(
            "https://jsonplaceholder.typicode.com/posts"
        );

        response.EnsureSuccessStatusCode();

        var json = await response.Content.ReadAsStringAsync();

        var posts = JsonSerializer.Deserialize<List<Post>>(
            json,
            new JsonSerializerOptions
            {
                PropertyNameCaseInsensitive = true
            }
        );

        return View(posts ?? new List<Post>());
    }

    // Appointment
    [HttpPost]
    public IActionResult Appointment(
        int doctorId,
        string patientName)
    {
        // Cookie
        Response.Cookies.Append(
            "PatientName",
            patientName
        );
```

```
        TempData["Message"] =
            $"Appointment request submitted for Doctor ID {doctorId}.";

        return RedirectToAction("Index");
    }

    // Error
    public IActionResult Error()
    {
        return View(new ErrorViewModel
        {
            RequestId = HttpContext.TraceIdentifier
        });
    }
  }
}
```

Views/_ViewImports.cshtml

```
@using HospitalAppointment
@using HospitalAppointment.Models
@using HospitalAppointment.Extensions

@addTagHelper *, Microsoft.AspNetCore.Mvc.TagHelpers
```

Views/_ViewStart.cshtml

```
@{
    Layout = null;
}
```

Views/Home/Index.cshtml

```
@model List<HospitalAppointment.Models.Doctor>

<!DOCTYPE html>
<html>
<head>

    <meta charset="utf-8" />

    <title>
        Hospital Appointment Management
    </title>

    <link rel="stylesheet"
```

```
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css">

</head>

<body>

<div class="container mt-5">

    <h1 class="text-center mb-4">
        Hospital Appointment Management System
    </h1>

    <hr />

    <h2>Doctor List</h2>

    <table class="table table-bordered table-striped mt-3">

        <thead class="table-dark">

            <tr>

                <th>Doctor ID</th>

                <th>Doctor Name</th>

                <th>Specialization</th>

                <th>Experience</th>

                <th>Consultation Fee</th>

                <th>Total Charge</th>

                <th>Details</th>

            </tr>

        </thead>
```

```
<tbody>

@foreach (var doctor in Model)
{
    <tr>

        <td>
            @doctor.DoctorId
        </td>

        <td>
            @doctor.DoctorName
        </td>

        <td>
            @doctor.Specialization
        </td>

        <td>
            @doctor.Experience Years
        </td>

        <td>
            ₹@doctor.ConsultationFee
        </td>

        <td>
            ₹@doctor.CalculateTotalCharge()
        </td>

        <td>

            <a href="/Home/Details?id=@doctor.DoctorId"
               class="btn btn-primary btn-sm">

                View Details

            </a>

        </td>
```

```
            </tr>
        }

        </tbody>

    </table>

    <hr />

    <h2>Patient Appointment</h2>

    <form method="post"
          action="/Home/Appointment">

        <div class="mb-3">

            <label class="form-label">
                Patient Name
            </label>

            <input type="text"
                   name="patientName"
                   class="form-control"
                   placeholder="Enter patient name"
                   required />

        </div>

        <div class="mb-3">

            <label class="form-label">
                Doctor ID
            </label>

            <input type="number"
                   name="doctorId"
                   class="form-control"
                   placeholder="Enter Doctor ID"
                   required />
```

```
        </div>

        <!-- Hidden Field -->

        <input type="hidden"
               name="doctorId"
               value="1" />

        <button type="submit"
                class="btn btn-success">

            Book Appointment

        </button>

    </form>

    <hr />

    <a href="/Home/Posts"
       class="btn btn-info">

        View Posts

    </a>

    @if (TempData["Message"] != null)
    {
        <div class="alert alert-success mt-3">

            @TempData["Message"]

        </div>
    }

</div>

</body>
</html>
```

Views/Home/Details.cshtml

```
@model HospitalAppointment.Models.Doctor

<!DOCTYPE html>
<html>
<head>

    <meta charset="utf-8" />

    <title>Doctor Details</title>

    <link rel="stylesheet"

href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstra
p.min.css">

</head>

<body>

<div class="container mt-5">

    <h1>
        Doctor Details
    </h1>

    <table class="table table-bordered mt-4">

        <tr>
            <th>Doctor ID</th>
            <td>@Model.DoctorId</td>
        </tr>

        <tr>
            <th>Doctor Name</th>
            <td>@Model.DoctorName</td>
        </tr>

        <tr>
            <th>Specialization</th>
            <td>@Model.Specialization</td>
        </tr>
```

```
        <tr>
            <th>Experience</th>
            <td>@Model.Experience Years</td>
        </tr>

        <tr>
            <th>Consultation Fee</th>
            <td>₹@Model.ConsultationFee</td>
        </tr>

        <tr>
            <th>Service Charge</th>
            <td>₹100</td>
        </tr>

        <tr class="table-primary">

            <th>
                Total Charge
            </th>

            <td>

                <b>
                    ₹@(Model.ConsultationFee + 100)
                </b>

            </td>

        </tr>

    </table>

    <a href="/Home/Index"
       class="btn btn-primary">

        Back

    </a>
```

```
</div>

</body>
</html>
```

Views/Home/Posts.cshtml

```
@model List<HospitalAppointment.Models.Post>

<!DOCTYPE html>
<html>
<head>

    <meta charset="utf-8" />

    <title>Posts</title>

    <link rel="stylesheet"

href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstra
p.min.css">

</head>

<body>

<div class="container mt-5">

    <h1>
        Posts
    </h1>

    <table class="table table-bordered table-striped mt-4">

        <thead class="table-dark">

            <tr>

                <th>
                    Post ID
                </th>
```

```
                <th>
                    Title
                </th>

            </tr>

        </thead>

        <tbody>

        @foreach (var post in Model)
        {
            <tr>

                <td>
                    @post.Id
                </td>

                <td>
                    @post.Title
                </td>

            </tr>
        }

        </tbody>

    </table>

    <a href="/Home/Index"
       class="btn btn-primary">

        Back

    </a>

</div>

</body>
</html>
```

Web .net cie 2 task

Views/Shared/Error.cshtml

```
@model HospitalAppointment.Models.ErrorViewModel

<!DOCTYPE html>
<html>
<head>

    <meta charset="utf-8" />

    <title>Error</title>

</head>

<body>

    <h1>Error</h1>

    <p>
        Something went wrong.
    </p>

    @if (Model.ShowRequestId)
    {
        <p>
            Request ID: @Model.RequestId
        </p>
    }

</body>
</html>
```

HospitalAppointment.csproj

```
<Project Sdk="Microsoft.NET.Sdk.Web">

  <PropertyGroup>

    <TargetFramework>net6.0</TargetFramework>

    <Nullable>enable</Nullable>

    <ImplicitUsings>enable</ImplicitUsings>

  </PropertyGroup>

</Project>
```

Web .net cie 2 task

Output

Hospital Appointment Management System

Doctor List

Doctor ID	Doctor Name	Specialization	Experience	Consultation Fee	Total Charge	Details
1	Dr. Amit Patel	Cardiology	10 Years	₹500	₹600	View Details
2	Dr. Neha Shah	Dermatology	7 Years	₹400	₹500	View Details
3	Dr. Raj Mehta	Neurology	12 Years	₹700	₹800	View Details

Patient Appointment

Patient Appointment

Patient Name

Enter patient name

Doctor ID

Enter Doctor ID

Book Appointment

View Posts

Appointment request submitted for Doctor ID 123.

Web .net cie 2 task

Posts

Post ID	Title
1	sunt aut facere repellat provident occaecati excepturi optio reprehenderit
2	qui est esse
3	ea molestias quasi exercitationem repellat qui ipsa sit aut
4	eum et est occaecati
5	nesciunt quas odio
6	dolorem eum magni eos aperiam quia
7	magnam facilis autem
8	dolorem dolore est ipsam
9	nesciunt iure omnis dolorem tempora et accusantium
10	optio molestias id quia eum
11	et ea vero quia laudantium autem
12	in quibusdam tempore odit est dolorem
93	beatae soluta recusandae
94	qui qui voluptates illo iste minima
95	id minus libero illum nam ad officiis
96	quaerat velit veniam amet cupiditate aut numquam ut sequi
97	quas fugiat ut perspiciatis vero provident
98	laboriosam dolor voluptates
99	temporibus sit alias delectus eligendi possimus magni
100	at nam consequatur ea labore ea harum

Back